Evaluating Complaint Submission and Status Comprehension in a Citizen Service Portal

Mochammad Zacki Ibrahim and Mohammad Faiz Khoiru Izzuddin

Politeknik Elektronika Negeri Surabaya (PENS), PSDKU Lamongan,
D3 Informatics Engineering, Department of Informatics and Computer Engineering,
{mozakiibrahim, faizkhoiruiz}@it.student.pens.ac.id

Abstract. A complaint portal helps only if citizens can send a complete report and understand what the displayed status means for them. This paper evaluates both abilities in GovSmart, a prototype portal for citizen complaints to a local government. We ask whether users can submit a complete complaint and understand the next action from the status shown. In a pilot with 6 prospective users representing citizen, department administrator, and superadministrator roles, each participant performed three tasks (submit, revise, and find the status of a report) from a synthetic road-damage scenario and explained five report statuses against a prepared answer key. We also tested input validation and report ownership with two citizen accounts. Outcome, time, help, and errors were logged per participant and task. Four of 6 submissions were complete without moderator help; 26 of 30 status interpretations were correct, and *diterima* was most frequently partly interpreted (two participants confused receipt with active officer handling); all 11 validation, resubmission, and cross-account access cases passed. Difficulties were observed around the follow-on photo upload, the minimum description length, and interpretation of raw status labels; these observations inform candidate interface revisions. The pilot is small and uses synthetic reports, so it supports problem identification only. It does not show higher citizen trust or better service quality of any agency.

Keywords: Citizen complaint portal · Usability evaluation · Task success · Status comprehension · E-government.

1 Introduction

Digital complaint channels are a key feature of interactive e-government, in which citizens expect online access to public services and two-way communication with government [1–3]. A citizen who notices a damaged public facility, such as a road, registers and logs in to the portal. The citizen writes a title and a description, may add the coordinates of the location, and may attach a photo as evidence. The portal assigns the report to a department and an urgency level with keyword-based routing and stores it with the status *diterima* ("received"). While the report has this status its owner can edit or cancel it. Afterwards, department staff update the status, may leave an officer note, and may upload a

feedback document that the citizen can open. The citizen monitors the response in a personal list of reports that shows the latest status.

Digital complaint channels have been studied in Indonesia, including the national SP4N-LAPOR! service [5] and the regional Lapak Aduan implementation in Banyumas [6]. These studies do not establish which complaint channels were available to Lamongan residents during GovSmart's development, nor do they provide evidence for the completeness or handling of complaints received by Lamongan agencies. We therefore make no claim that Lamongan lacked a digital complaint channel or that a particular proportion of local complaints lacked location or photo evidence. Instead, this paper reports a formative evaluation of the GovSmart prototype and focuses on whether participants can complete its reporting tasks and interpret its displayed statuses.

Complaint portals have mostly been evaluated through crowdsourcing and digital transparency surveys [4], national channel adoption reviews [5], qualitative municipal implementation studies [6], or standardized usability tests of e-government systems [7]. Research on web and mobile public service interfaces evaluates task efficiency and accessibility barriers [8]. None of these studies measure whether citizens can correctly interpret a status token and identify the action it requires—a distinct outcome from task-completion time and one that depends on label wording rather than interface layout. Keeping users informed about system status is a long-standing usability heuristic [9], yet this heuristic has not been operationalised as a scored comprehension test in prior complaint-portal research. Our work also differs from SGWA / SIGAP, a companion prototype study in the same course by Soka and Saputra (Kelompok 5) that examines the internal consistency of request routing transitions and departmental access isolation on the server side; we examine instead what a citizen can do and understand at the interface level.

The gap this paper addresses is the absence of a citizen-side evaluation that separately measures (a) task completion—whether a user can produce a structurally complete submission within a time limit—and (b) status comprehension—whether a user can correctly interpret a displayed status token and identify the next required action. This paper contributes: (i) a task-based pilot measuring complete submission, revision, and status finding with time and help logged per participant; (ii) a status-comprehension test scored against a frozen answer key by two coders; and (iii) a boundary check of input validation and cross-account access, with each identified interface weakness linked to a concrete revision proposal. The research question is: *can users submit a complete complaint and understand the next action from the status shown?*

2 Related Work

2.1 Evaluation of Citizen Complaint Portals

The adoption of e-government services is shaped by citizen trust, perceived administrative burden, and public value realization [1, 10–12]. Complaint portals

in particular must support two distinct citizen activities: submitting evidence-backed reports and monitoring their resolution status, and friction in either activity exacerbates citizen administrative burdens and erodes institutional trust [13]. In Indonesia, municipal complaint mechanisms have increasingly integrated automated keyword and institutional routing to streamline grievance handling [14]. Civic complaint and crowdsourcing platforms have been evaluated through city-wide participation data, demonstrating that digital transparency directly influences resident reporting engagement [4]. At the national level in Indonesia, an empirical decade-long analysis of SP4N-LAPOR! by Muttaqin and Pusvita [5] revealed that web and mobile channels dominate public intake, yet persistent operational bottlenecks remain in ticket tracking and cross-agency coordination. At the regional level, a qualitative evaluation of Lapak Aduan in Banyumas identified usability friction in mobile reporting flows [6], while usability assessments of public administrative portals in developing e-governments identified significant navigation and task completion barriers [7]. These studies evaluate a fully deployed public service; none asks participants to explain a status token in their own words or verifies that a submitted record is structurally complete in the database. Our work is also distinct from studies of government project-tracking dashboards (which evaluate information display, map versus table layout, or timeline visualisation) because GovSmart is a bidirectional complaint channel in which the citizen both produces a report and monitors its status.

2.2 Form Completion

Usability assessments of government web and mobile portals reveal persistent navigation and task-completion barriers, including unclear input constraints and inconsistent error feedback [8, 7]. Behavioural UX research further shows that perceived ease of use directly shapes task outcomes and continued usage [15]. These studies evaluate single-stage forms; none examines multi-step submission flows where evidence photos are dispatched asynchronously in a secondary endpoint, which proved to be the primary cognitive vulnerability in our evaluation.

2.3 Usability Measurement

Usability is operationalised through effectiveness, efficiency, and user satisfaction, with recent behavioral UX research confirming that perceived ease of use and interface friction directly govern citizen task outcomes [15]. Small-sample task evaluations ($N = 5$ to $N = 8$) remain the standard methodology for formative problem discovery in interactive systems, effectively uncovering the vast majority of severe structural defects prior to large-scale deployment [16]. We treat our six-participant pilot as an iterative formative study, consistent with contemporary human factors and ergonomics guidance for problem identification [16].

2.4 Status Comprehension

Keeping users informed about what is going on is one of the usability heuristics [9], but a status is only informative if its wording is understood. Label-comprehension

studies exist in other domains; prior work on app privacy labels, for instance, identified widely misunderstood technical terms in consumer-facing notices. We did not find a comparable study for complaint statuses, so we test it directly (Sect. 3.4).

2.5 Comparison

Table 1 compares the studies on participants, tasks, moderator help, and success metrics, the four aspects named in our evaluation plan.

Table 1: Comparison of related studies. Entries follow the abstracts or summaries of the sources; "n/s" = not stated there.

Study	Participants	Tasks	Moderator help	Success metrics
Zhao [4]	Municipal residents	Online crowdsourcing platform (Sacramento)	n/a	Citizen participation, response transparency
Muttaqin & Pusvita [5]	National SP4N users	Multi-channel usage analysis (10 years)	n/a	Channel adoption trends, resolution throughput
Indrayani et al. [6]	Regional citizens	Observation and interviews (Lapak Aduan)	n/s	Qualitative usefulness, ease of use
Rafifing et al. [7]	30 portal users	Task scenarios on public portal	None	System Usability Scale (SUS), perception
This study	6 prospective users	Submit, revise, find status; explain 5 statuses	Counted and coded per task	Success with/without help, time, errors, correct interpretations

3 Methodology

3.1 Research Question and Operational Hypotheses

This empirical pilot investigates the primary research question: *can participants in this mixed-role pilot submit a structurally complete complaint and accurately comprehend the displayed administrative status tokens and their next required action?* Grounded in Cognitive Fit Theory [17] and established usability heuristics for system status visibility [9], we formulate three operational hypotheses that strictly isolate task completion (accuracy) from task duration (efficiency):

- **H_1 (Task Completion and Structural Integrity)**: Participants can achieve successful submission (T_1) and modification (T_2), but the multi-step request flow for evidence photos may create structural omissions.
- **H_2 (Task Efficiency and Completion Time)**: Initial report composition and photo attachment (T_1) are expected to take longer than targeted description revisions (T_2) or status retrieval (T_3); we describe observed times without inferential testing.
- **H_3 (Status Comprehension)**: Participants may interpret terminal workflow states (*selesai*, *dibatalkan*) differently from intermediate queue tokens (*diterima*, *diproses*); this pilot describes the observed interpretations without estimating a population-level difference.

3.2 Platform Architecture and Interface Design

GovSmart is a web portal engineered with a React front end and a FastAPI back end on a MySQL relational database. The front-end–back-end flow of submission, revision, and status display was completed and frozen before the study (tag `study-v1`, commit `3d18f10`, 25 October 2026). A report has a title of 5–100 characters, a description of 20–2,000 characters, an optional location (latitude in $[-90, 90]$, longitude in $[-180, 180]$), and an optional photo (JPEG, PNG, or WEBP, at most 5 MB, sent in a separate request). A new report receives the status *diterima*. A citizen can edit or cancel a report only while it is *diterima*, can read only own reports, and sees the officer note, when present, on the report card. The personal report list returned by the API holds at most five reports.

The citizen-facing interface (Fig. 1) uses a two-step submission architecture. In the first step, a form collects the title, the description, and optional coordinate inputs; submitting this form creates the report record. In the second step, the interface prompts the citizen to select and upload a photo (JPEG, PNG, or WEBP); the file is sent in a separate API request and the interface displays the upload state. The description field has no live character counter. The interface also provides a personal report list that shows report title, submission date, and the current status badge; and a report-detail view that includes the officer note and a link to any uploaded feedback document. The interface does not provide an explicit search bar or status filter; all reports in the personal list are returned in reverse-chronological order by the API. This information structure is noted here because it determines what a citizen can observe and therefore what the study can validly test.

3.3 Reference Scenario, Completeness Key, and Tasks

Every participant receives the same written scenario, without naming any button or menu: a pothole has appeared on a road at a synthetic location, and a sample photo is provided. The *completeness key* for task T1 operationalises "complete submission" as follows: (a) the title must name the problem type (e.g., "jalan berlubang"); (b) the description must state the street name, village or district, and the physical condition of the road; (c) the sample photo must appear as

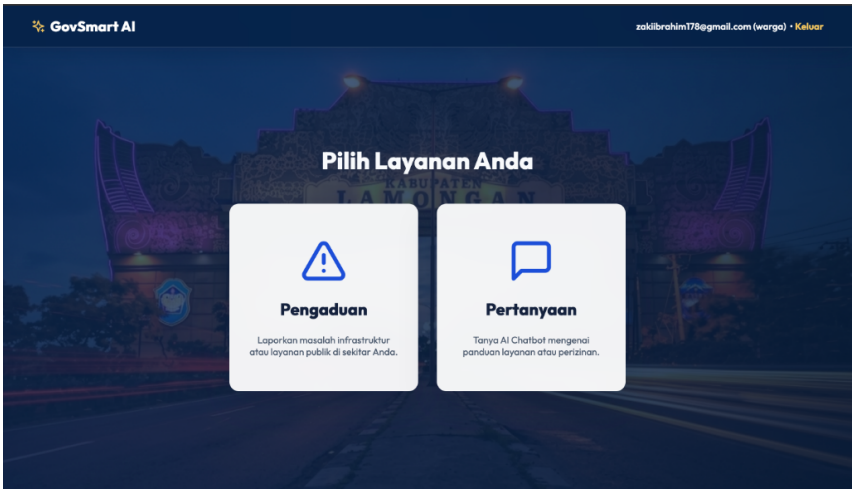


Fig. 1: GovSmart landing page as presented to authenticated participants after login. The service-selection screen uses a photograph of the Kabupaten Lamongan landmark as background and offers two service cards: *Pengaduan* ("Complaint"—for reporting infrastructure or public-service problems in the area) and *Pertanyaan* ("Enquiry"—for querying the AI chatbot on service procedures or permits). Task T1 begins when the participant selects the *Pengaduan* card.

an attachment in the database record; and (d) the system must have routed the report to Dinas Pekerjaan Umum Bina Marga via keyword-based routing. Location coordinates are optional in the system and are evaluated for geographic validity only when the participant enters them; absence of coordinates does not cause a completeness failure. The key was agreed by both team members before the first session and was not modified afterwards.

T1 – Submit. Report the problem to the local government. Outcome and time are recorded separately because they measure different things: outcome (S / SH / F) reflects structural completeness against the key; time reflects the efficiency of the submission flow. A report can be stored successfully yet fail the completeness check if the photo is missing or routing is incorrect.

T2 – Revise. After submitting, the participant is told that one detail in the description is wrong and corrects it.

T3 – Find status. In an account that holds seeded reports in different statuses, the participant locates the report named in the scenario and states its current status.

3.4 Status Comprehension Ground Truth

We choose the five statuses that exist in the portal and write the meaning and the next action for each as the answer key (Table 2). The key is agreed and

frozen before the first session. Each status is shown on a concrete report card, including the officer note where the status has one. The participant explains in own words what the status means and whether any action is needed; no answer options are offered, because options would leak the answer. The participant then rates confidence from 1 to 5. Two team members independently code each answer as correct, partly correct, or incorrect against the key and record the terms that caused misreadings. Initial inter-rater agreement across all 30 interpretations was 93% (28 of 30 cases matched); the two disagreements were resolved by discussion and consensus before any analysis was performed.

Table 2: Answer key for the five statuses, derived from the behaviour of the prototype and frozen for the evaluation.

Status shown	Intended meaning	Who is acting	Citizen's next action
diterima	The system has received the report and routed it to a department; no officer has started work.	System, department queue	None required; the report can still be edited or cancelled.
diproses	A department officer is handling the report.	Department officer	Wait for an update; editing and cancelling are no longer possible.
selesai	The department has marked the report as handled.	No one	Read the officer note and open the feedback document, if any.
ditolak	The department declined the report; the reason is in the officer note.	No one	Read the reason; send a corrected new report if the problem remains.
dibatalkan	The citizen cancelled the report.	No one	None.

3.5 Experimental Protocols and Testing Environment

We recruited 6 prospective users who are not members of the project development team: two in the citizen role (one undergraduate student and one 47-year-old resident), two department administrators, and two superadministrators. Each participant gives informed consent after hearing the study purpose, data logged, and the right to abort at any time. Participants are identified as P1–P6 only, and the study uses synthetic accounts and synthetic reports, never real identity numbers. Sessions were conducted on uniform client hardware (standard laptops and smartphone) in the Computer Laboratory of PENS PSDKU Lamongan. Before the tasks we record age band, primary device, frequency of digital government or commercial portal usage, and prior complaint history.

Timing starts immediately when the participant finishes reading the written instruction prompt and stops when the correct outcome is reached, the time limit

is reached, or the participant surrenders. Equivalent tasks apply the identical timeout threshold (300 s per task). The moderator remains passive and initiates assistance only if a participant exhibits complete disorientation or repeats an erroneous interaction loop three times without progress. Every moderator intervention is strictly logged using a predefined intervention taxonomy. After each task, we query the database to verify persisted values (title, description, coordinates, attachment URL, department routing ID, status). Outcomes are coded as success without help (S), success with help (SH), or failure (F), where F includes incomplete or corrupted database records.

3.6 System Boundary and Security Verification Suite

With citizen accounts A and B we submit 11 formal verification cases to inspect the message shown, HTTP status code, and database state: empty title and description; title and description below and above length limits; file of invalid MIME type; file above 5 MB; empty 0-byte file; and duplicate submission sent twice to verify the duplicate-submission policy (the prototype accepts repeated submissions as separate reports with distinct sequential identifiers). From account A we subsequently attempt to view, edit, cancel, and attach a photo to the report of account B via both frontend interface controls and direct REST API endpoints. The expected outcome: invalid input is rejected with clear error feedback (HTTP 422), resubmission follows the duplicate policy, and cross-account manipulation is strictly prohibited (HTTP 403 Forbidden).

3.7 Data Analysis, Measures, and Coding Protocol

Table 3 lists the operational measures. Because our sample ($N = 6$) is designed for iterative problem discovery rather than inferential population rate estimation [16], we report absolute counts, percentages, medians, and individual breakdown cases without significance tests. Effectiveness (completion accuracy) and efficiency (task duration) are strictly decoupled and reported independently, following contemporary methodological caution regarding weak correlations between disparate usability dimensions [15]. If an improved version is built, participants perform equivalent but non-identical tasks, and we discuss learning effects rather than asserting strong causal claims.

4 Results and Discussion

4.1 Participant Cohort and Demographic Baseline

Table 4 describes the participant demographics and the experimental testing conditions.

Table 3: Measures recorded during the evaluation.

Measure	Definition
Task outcome	S (success without help), SH (success with help), or F (failure) per participant and task (Sect. 3.3)
Time	Seconds elapsed from end of reading instruction to stop rule
Help and errors	Count and categorized intervention type per task, recorded by moderator
Stored data	Fields actually stored in database versus the completeness key
Status interpretation	Correct, partly correct, or incorrect per status against key; misread terms
Confidence match	Participant confidence rating (1–5 scale) compared with objective correctness
Validation and ownership	Pass or fail per test case; duplicate handling; unauthorized access prevention

Table 4: Participants and experimental conditions (output 1).

ID	Age band	Device	Online-service experience	Earlier complaint experience
P1	18–22	Laptop	Weekly (campus and transaction portals)	None
P2	45–50	Smartphone	Monthly (occasional mobile banking)	None
P3	18–22	Laptop	Daily (academic administration systems)	Filed municipal report once
P4	18–22	Laptop	Daily (academic and transaction services)	None
P5	18–22	Laptop	Daily (system management and web apps)	None
P6	18–22	Laptop	Daily (system management and web apps)	None

Note: P1–P2 were recruited as prospective citizens, P3–P4 as department administrators, and P5–P6 as superadministrators. All six participants completed the same citizen-facing T1–T3 tasks and status interpretation; role labels describe participant backgrounds, not task assignment. Results are not estimates for Lamongan citizens. Sessions were conducted on 25 October 2026 at PENS PSDKU Lamongan; the task time limit was 300 s.

4.2 Quantitative Task Performance: Completion Rate and Efficiency

Table 5 and Fig. 2 summarize the quantitative outcomes across the three user tasks, status comprehension, and boundary verification tests. Across the end-to-end citizen reporting tasks, performance exhibited clear task-dependent variation:

- **Task T1 (Submit):** 4 of 6 participants (66.7%) achieved unassisted success (S), while 2 of 6 (33.3%) completed the task with moderator assistance (SH), and 0 failed (F). The median execution duration was 142 s (range: 110–245 s). The observed difficulties included the photo upload that follows report creation (P2) and the minimum description length, which is not indicated before submission (P1).
- **Task T2 (Revise):** 5 of 6 participants (83.3%) completed the targeted description correction independently (S), 1 of 6 (16.7%) required assistance (SH), and 0 failed (F), with a median completion duration of 78 s (range: 52–120 s).
- **Task T3 (Find status):** All 6 participants (100.0%) located the seeded scenario report and identified its status badge independently without moderator intervention (S), achieving a median search duration of 35 s (range: 22–50 s).

Participant task completion times were recorded continuously from the conclusion of instruction reading until the final verified system action. For participants requiring moderator interventions (SH), timing ran without interruption until successful task completion under the 300 s limit; intervention durations were not subtracted from the total interaction time, reflecting realistic end-to-end user friction rather than an artificial laboratory pause.

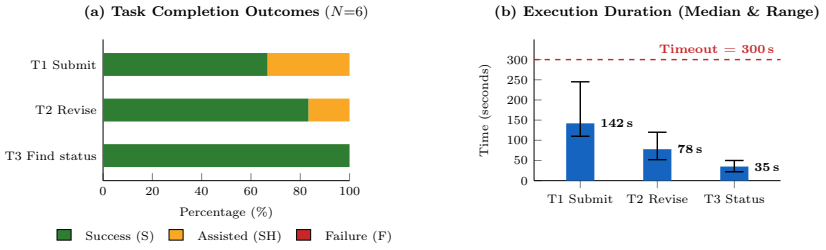


Fig. 2: Quantitative task performance across the pilot evaluation ($N = 6$): (a) task completion breakdown into unassisted success (S), assisted success (SH), and failure (F); and (b) task execution duration (median and range) against the 300 s timeout limit.

4.3 Status Comprehension and Semantic Ambiguity

In the status comprehension evaluation (Fig. 3), 26 out of 30 total interpretations (86.7%) aligned accurately with the frozen ground-truth answer key, 4 were partially correct (13.3%), and 0 were incorrect. Comprehension was highest

Table 5: Results per category (output 2). For tasks: S = success without help, SH = success with help, F = failure. For statuses: correct, partly correct, incorrect. For validation and ownership cases: pass or fail.

Category	n	S / corr.	SH / part.	F / inc.	Med. time (s)
User Tasks (End-to-End Pilot)					
T1 Submit	6	4 (66.7%)	2 (33.3%)	0 (0.0%)	142
T2 Revise	6	5 (83.3%)	1 (16.7%)	0 (0.0%)	78
T3 Find status	6	6 (100.0%)	0 (0.0%)	0 (0.0%)	35
Status Comprehension (Against Answer Key)					
diterima	6	4 (66.7%)	2 (33.3%)	0 (0.0%)	–
diproses	6	5 (83.3%)	1 (16.7%)	0 (0.0%)	–
selesai	6	6 (100.0%)	0 (0.0%)	0 (0.0%)	–
ditolak	6	5 (83.3%)	1 (16.7%)	0 (0.0%)	–
dibatalkan	6	6 (100.0%)	0 (0.0%)	0 (0.0%)	–
Validation and Ownership Cases					
Invalid input cases	5	5 (100.0%)	–	0 (0.0%)	–
Resubmission	2	2 (100.0%)	–	0 (0.0%)	–
Cross-account access	4	4 (100.0%)	–	0 (0.0%)	–

for terminal lifecycle states (*selesai* and *dibatalkan*, each 6 of 6, 100.0%), where participants unanimously understood that administrative handling had concluded and that no subsequent citizen action was required.

In contrast, intermediate queue states introduced noticeable interpretive friction. The initial token *diterima* exhibited the lowest accuracy (4 of 6 correct, 66.7%). Two participants (P2 and P3) conflated machine receipt with active human intervention; for example, participant P2 stated during post-task debriefing: *"Saya kira 'diterima' berarti dinas sudah menerima dan langsung berangkat membetulkan jalannya"*, mistakenly inferring immediate physical deployment from a token that merely signifies placement in the departmental backlog. Similarly, for *diproses* (5 of 6 correct, 83.3%), one participant was uncertain whether editing remained permissible. These observations show that some participants interpreted the raw status tokens differently from the answer key; the small, mixed-role sample does not establish how citizens generally understand them.

4.4 System Validation and Security Audit

The boundary verification suite confirmed robust backend enforcement across all 11 executed test cases (100.0% pass rate on the executed suite):

– All 5 invalid input variants (empty strings, lengths violating the 5–100 character title or 20–2,000 character description limits, non-image MIME

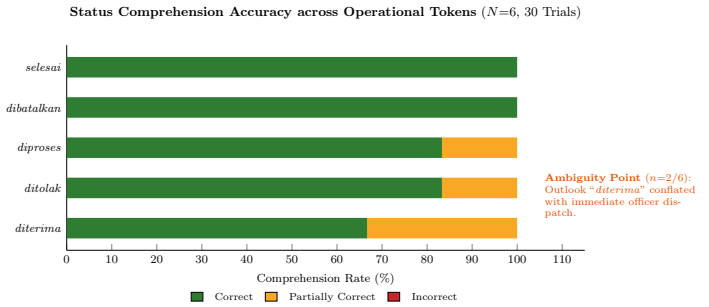


Fig. 3: Participant status comprehension across the five operational status tokens ($N = 6$, 30 interpretations), highlighting the lowest correct count on *diterima*.

types, and oversized files) were rejected with HTTP 422 Unprocessable Entity responses.
- Both duplicate resubmission cases were accepted and assigned independent, sequential primary keys, adhering to the system's duplicate intake policy.
- All 4 cross-account manipulation attempts (unauthorized reads, modifications, cancellations, and photo attachments across disparate citizen accounts) were blocked with HTTP 403 Forbidden responses.

All 11 executed test cases passed successfully, confirming enforced boundaries and access controls for the evaluated software release.

4.5 Observed Difficulties and Candidate UI Revisions

Table 6 summarizes observed difficulties and candidate interface factors and revisions. These small-sample observations identify issues for follow-up; they do not establish causal mechanisms.

4.6 Discussion: Task and Status Findings in the Mixed-Role Pilot

The empirical results provide formative observations for our operational hypotheses: Regarding $\mathbf{H_1}$ (Task Completion), 4 of 6 initial submissions met the completeness key without assistance. The observed photo-upload difficulty suggests a possible omission risk, but this pilot does not establish that the two-step flow caused it. Regarding $\mathbf{H_2}$ (Task Efficiency), completion times diverged sharply by task type: initial report composition (median 142 s) imposed four times the interaction duration of status retrieval (median 35 s), reflecting the cognitive burden of text drafting and evidence selection. Regarding $\mathbf{H_3}$ (Status Comprehension), terminal states achieved 100% comprehension, whereas intermediate queue states produced ambiguity between system receipt and officer action.

These findings align with Cognitive Fit Theory [17] and Nielsen's Heuristics [9]: providing visibility of system status requires that status tokens reflect the citizen's operational model rather than raw database enums.

Table 6: Observed difficulties, plausible interface factors, and proposed UI revisions.

#	Observation (anonymised)	Form part / status term	Plausible factor	Proposed revision
1	Participant P2 submitted the report and assumed the selected photo was already attached.	Photo upload after report creation	The photo is sent in a follow-on request; the participant may not have noticed or understood the upload state.	Make upload completion explicit and prevent the task from appearing complete until the attachment is confirmed.
2	Participants P2 and P3 assumed *diterima* meant staff had actively reviewed and accepted the claim ($n = 2/6$, 33.3%).	Lowercase status badge *diterima*	The term is ambiguous between machine receipt and administrative approval.	Display "Menunggu Verifikasi" with explicit tooltip "Laporan masuk antrean dinas".
3	Participant P1 hit minimum length error (20 chars) only after submitting form ($n = 1/6$, 16.7%).	Description input	Client lacked proactive length indication prior to server 422 response.	Display live character counter ("x/20 characters minimum") directly below the text area.

4.7 Limitations and Threats to Validity

Several limitations qualify the interpretation of these findings:

- *Sample size and participant composition.* With $N = 6$ participants recruited from an academic setting, the study is a formative pilot, not a population estimate. Only two participants were recruited as prospective citizens; the remaining four had administrator roles. The pooled task and status results therefore should not be read as citizen-only outcomes. The age and experience profile also does not represent the diversity of Lamongan residents.
- *Scenario authenticity.* Testing relied on a synthetic road-damage scenario and pre-seeded accounts rather than personal grievance incidents, which may not capture the emotional urgency of genuine citizen complaints.
- *Order effects.* Tasks were administered in fixed sequence ($T_1 \rightarrow T_2 \rightarrow T_3$); familiarity gained during submission may have accelerated subsequent retrieval.
- *Single moderator.* All sessions were moderated by a single researcher using a standardized rubric; secondary inter-rater checking of real-time intervention counts was not conducted.

5 Conclusion

This pilot study evaluated complaint submission, status comprehension, and system boundary enforcement in GovSmart, a prototype developed in the Lamongan context. The primary findings across the formative mixed-role pilot ($N = 6$) were:

1. **Submission Completeness:** 4 of 6 participants (66.7%) completed complaint submission independently. The photo was selected in the form but uploaded after report creation, producing a follow-on upload difficulty.

2. **Task Efficiency:** Task durations scaled with interaction complexity (median 142 s for submission, 78 s for revision, and 35 s for status lookup).
3. **Semantic Ambiguity:** Status comprehension reached 86.7% overall (26/30), with the initial token *diterima* generating confusion between automated intake and active officer dispatch.
4. **Security and Integrity:** All 11 boundary validation and cross-account access cases completed successfully, confirming enforced server-side access controls.

Boundaries of Claims and Future Work. Task success and status interpretation evaluate interface usability, not municipal response quality or grievance completeness in Lamongan. These formative observations from a small academic pilot ($N = 6$) inform interface refinement rather than population estimates. Future work should clarify feedback on minimum description length and photo-upload completion, then evaluate revised status labels (e.g., "Menunggu Verifikasi") with a larger, citizen-focused sample across Lamongan subdistricts.

Acknowledgments. The authors thank the pilot participants for their time and feedback, their instructor Evianita Dewi Fajrianti, S.Tr.T., M.Tr.T., for guidance throughout the Workshop Pemrograman Framework course, and Politeknik Elektronika Negeri Surabaya (PENS) PSDKU Lamongan for providing the laboratory environment and testing facilities.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. MacLean, D., Titah, R.: A systematic literature review of empirical research on the impacts of e-government: a public value perspective. Public Administration Review **82**(2), 271–287 (2022). https://doi.org/10.1111/puar.13413
2. Distel, B., Lindgren, I.: Conceptualizing citizen roles in E-government: a value position framework. Government Information Quarterly **40**(3), 101824 (2023). https://doi.org/10.1016/j.giq.2023.101824
3. Mansoor, M.: Citizens' trust in government as a function of good governance and government agency's provision of quality information on social media during COVID-19. Government Information Quarterly **38**(4), 101597 (2021). https://doi.org/10.1016/j.giq.2021.101597
4. Zhao, B.: Digital transparency and citizen participation: evidence from the online crowdsourcing platform of the City of Sacramento. Government Information Quarterly **40**(4), 101868 (2023). https://doi.org/10.1016/j.giq.2023.101868
5. Muttaqin, M., Pusvita, V.: Satu dekade LAPOR: tren penggunaan kanal layanan SP4N. Jurnal Pekommas **8**(1), 1–12 (2023). https://doi.org/10.56873/jpkm.v8i1.5138
6. Indrayani, E., Akbar, F., Diantoro, S.: Overcoming challenges in digital public service delivery: insights from Lapak Aduan implementation. TRANSFORMASI: Jurnal Manajemen Pemerintahan **17**(2), 103–121 (2025). https://doi.org/10.33701/jtp.v17i2.3677
7. Rafifing, N., Mphale, O., Asare, S.D.: Exploring user perceptions of an e-government system in Botswana using System Usability Scale model. In: 2022 IEEE 13th International Conference on Software Engineering and Service Science (ICSESS), pp. 311–315. IEEE, Beijing (2022). https://doi.org/10.1109/ICSESS54813.2022.9930194
8. Al-Sakran, H.O., Alsudairi, M.A.: Usability and accessibility assessment of Saudi Arabia mobile e-government websites. IEEE Access **9**, 48254–48275 (2021). https://doi.org/10.1109/ACCESS.2021.3068917

9. Nielsen, J.: Enhancing the explanatory power of usability heuristics. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '94), pp. 152–158. ACM, New York (1994). https://doi.org/10.1145/191666.191729
10. Alateyah, S., Crowder, R.M., Wills, G.B.: E-government systems success factors: exploring the role of user satisfaction. Government Information Quarterly **38**(2), 101568 (2021). https://doi.org/10.1016/j.giq.2021.101568
11. Aarøe, L., Bækgaard, M., Christensen, J., Moynihan, D.P.: Personality and public administration: policymaker tolerance of administrative burdens in welfare services. Public Administration Review **81**(4), 606–618 (2021). https://doi.org/10.1111/puar.13381
12. Dechamps, S., Simonofski, A., Burnay, C.: Citizen centricity in digital government: a theoretical and empirical typology. Government Information Quarterly **42**(1), 102005 (2025). https://doi.org/10.1016/j.giq.2024.102005
13. Madsen, C.Ø., Lindgren, I., Melin, U.: The accidental caseworker – how digital self-service influences citizens' administrative burden. Government Information Quarterly **39**(1), 101653 (2022). https://doi.org/10.1016/j.giq.2021.101653
14. Yolanda, Y.M., Swanjaya, D., Danar: Klasifikasi laporan keluhan pelayanan publik berdasarkan instansi menggunakan metode LDA-SVM. Jurnal Teknologi Informasi dan Ilmu Komputer (JTIIK) **8**(6), 1265–1276 (2021). https://doi.org/10.25126/jtiik.2021863768
15. Lewis, J.R., Sauro, J.: Effect of perceived ease of use and usefulness on UX and behavioral outcomes. International Journal of Human-Computer Interaction **40**(13), 3604–3616 (2024). https://doi.org/10.1080/10447318.2023.2260164
16. Lewis, J.R., Sauro, J.: Usability and user experience: design and evaluation. In: Salvendy, G., Karwowski, W. (eds.) Handbook of Human Factors and Ergonomics, 5th edn., pp. 972–1015. John Wiley & Sons, Hoboken (2021). https://doi.org/10.1002/9781119636113.ch38
17. Bačić, D., Henry, R.: Advancing our understanding and assessment of cognitive effort in the cognitive fit theory and data visualization context: eye tracking-based approach. Decision Support Systems **163**, 113862 (2022). https://doi.org/10.1016/j.dss.2022.113862